

Lab 3 (CS201)

1. Trajectories of an autonomous system: Basics Consider the autonomous system,

$$\dot{x} = \sin(x).$$

Integrate the ODE numerically using Euler's method with the following initial conditions, $x(0) = \pm 1, \pm 2$. Plot the four trajectories on the same graph. Comment on the asymptotic ($t \rightarrow \infty$) nature of the solutions and explain the results physically.

2. Confirming bifurcation from trajectories: Consider the autonomous systems,

- (a) $\dot{x} = r - x^2$. This system undergoes a saddle-node bifurcation at $r = 0$. Using Euler's method, integrate the equation numerically and show that the bifurcation takes place by analyzing the trajectories for $r = -1.1, 0, 1.1$. Choose initial conditions carefully. Using matplotlib (or similar) to make a slider whereby you can adjust the value of r and see the trajectories changing in real time.
- (b) $\dot{x} = rx - x^2$. This system undergoes a transcritical bifurcation at $r = 0$. Do the same exercise as the previous part.

3. Duffing Oscillator with low acceleration Consider the system,

$$\dot{x} = r - \alpha x - \beta x^3$$

Analyse the fixed points, stability and possible bifurcations of this system, with r being the tunable parameter. Simulate this system similar to the preceding questions. Note that there can be different combinations of the ranges of the parameters ($[\alpha > 0, \beta > 0]$, $[\alpha > 0, \beta < 0]$ etc) In addition, this system (which is motivated by the model of a forced oscillator in a viscous medium) is encountered in many areas of science and engineering and you should read at least qualitatively about this model. This information is readily available, for example, on Wikipedia. For example, what is the origin of the x^3 term?